

Education

The University of Texas at Austin — Austin, TX

Bachelor of Science: *Electrical and Computer Engineering*

Coursework: *Distributed Systems (Grad) | Systems for ML (Grad) | Multithreading & GPUs | Operating Systems | Computer Architecture*

Aug 2023 — May 2027

Publications

SOSP ‘26 (published) · EnerTune · “Beyond Utilization: Energy-Conscious GPU Sharing for Inference Serving”

https://dl.acm.org/doi/10.1145/3830418.3843913 | github.com/UT-SysML/EnerTune

3rd author

Experience

UT Austin Research · Undergraduate Research Assistant — Austin, TX

Machine Learning for Systems

• Building a harness streaming live vLLM stats to an LLM agent that tunes the request router, vLLM’s load balancer across replicas

Systems for Machine Learning

• Rebuilt 4 state-of-the-art GPU-sharing inference serving systems (GPUlets, Usher, FGD, ParvaGPU) to benchmark EnerTune

• Debugged and ported each baseline’s research prototype to run on our 16-A100 cluster, enabling the 1.4–2.3× lower energy result

• Profiled baselines to show EnerTune’s analytical model cuts profiling time 7.3× on average, up to 17.3×, versus prior systems

Computer Architecture and FPGAs

• Built an instruction-accurate RISC-V simulator in C++ so the team could test the “Primate” compiler without slow FPGA synthesis

• Designed I/O instruction handling and compiler-checking tools that sped up developing the hardware and software side by side

Mar 2024 — Current

Aug 2026 — Ongoing

Aug 2025 — May 2026

Mar 2024 — Dec 2024

Caterpillar Inc. · Software Engineering — Peoria, IL

Software Intern: Data Acquisition and Analytics Team

• Building a tool attributing cellular data costs to 20,000+ Caterpillar machines, raising cost accuracy 34% on average (up to 55%)

• Processing a 100 GB daily load from a 30–50 PB data pipeline to replace whole-SIM-bill accounting with per-megabyte attribution

• Deploying a nightly AWS Lambda that invokes carrier APIs to locate each machine, aggregate usage, and calculate its actual cost

• Assembling an ASP.NET dashboard with cross-source SQL joins and caching, giving every engineering team real-time cost visibility

Systems Intern: System Diagnostics Team

• Wrote hardware-in-the-loop tests covering 30 I/O types (PWM, analog, and digital) to set a baseline for the whole diagnostic suite

• Translated hardware requirements into embedded test modules so every new feature automatically gets regression-test coverage

• Automated key steps in the firmware build process used daily by a team of 15+ engineers, cutting their configuration time by 10%

Embedded Intern: Platform Software Team

• Designed hardware-abstraction (HAL) and API layers for engine control modules (NVM, analog, and digital I/O), now in production

• Developed firmware components in C and Python that cut new I/O support time by 15%, verified via .elf memory-layout checks

May 2024 — Current

Summer 2026, Part-Time through 2026–27

Summer 2025, Part-Time through 2025–26

Summer 2024, Part-Time through 2024–25

Longhorn Racing (Formula SAE) · Battery Management Developer — Austin, TX

Embedded Systems

• Led the design of the car’s battery monitoring system; wrote RTOS firmware modules handling drivers, I/O, and scheduling

• Built the State-of-Charge algorithm that improved battery range estimates, cutting the driving-range margin of error to under 5%

Aug 2023 — May 2025

Projects

Energy-Efficient MLLM Serving: Group Project, Systems for ML Grad Class (CUDA, vLLM)

Profiled multimodal LLM serving per phase on an A100, then 1,080 SM-count configurations for throughput, power and energy

Found decode takes 89% of runtime at batch size 1 while averaging only 14% SM throughput and 23% DRAM throughput

Measured 13× lower energy per request when batching output-heavy jobs, but only 2× on input-heavy ones due to encoder cost

mlm-energy.github.io/MLLM-Energy

Concurrent Hash Tables & Radix Trees: Multi-threaded Data Structure Benchmarking (C/C++)

Built 7 concurrent data structures in C++ for the prefix-cache lookups that bottleneck multi-threaded LLM serving engines

Achieved a 13× throughput gain on the same radix-tree structure by improving thread coordination, peaking at 88.36 million ops/sec

Found 40× p99 latency gaps between hash tables on a real ShareGPT trace, so workload shape should drive structure choice

code-tomato.github.io/CHT_Radix

Additional

Skills: C/C++, CUDA, Python, SQL, Bash | vLLM, gdb, QEMU, Docker, Git | AWS (Lambda), Oracle, MySQL | .NET, React, HTML/CSS, Tailwind

Awards: Samsung Scholarship, Santander Scholarship, Albert and Jessie Cudlipp Scholarship

Organizations & Leadership: Students4Hope (Founder), SEO USA, Ron Brown Scholar, ColorStack, NSBE, ABCS, OCCM